DESIGN LAB ASSIGNMENT

Course File Management System

Use Case Diagram • DFD • Sequence Diagram • State Transition Diagram

Group-2

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1. Use Case Diagram

Task 1.1: Primary Actors

- **Admin:** Manages system setup, course allocations, and user accounts.
- **Faculty:** Submits, manages, edits, and locks assigned course files and documents.
- **Course Coordinator:** Reviews submissions, checks completeness, consolidates content, and generates final course files.
- **AI System:** Performs automated document analysis and validation support.

Task 1.2: Main Use Cases per Actor

Admin

- View Dashboard / Login
- Manage Courses & Sections
- Assign Faculty
- Assign Course Coordinator
- Manage Users

Faculty

- Login
- View My Courses
- Upload Document
- Manage Submissions
- Edit / Replace / Delete
- Lock Submission

Course Coordinator

- View Dashboard
- Check Completeness
- Inform / Notify Faculty
- Consolidate Content
- Edit / Accept Submissions
- Generate Final Course File

AI System

- Analyze Documents

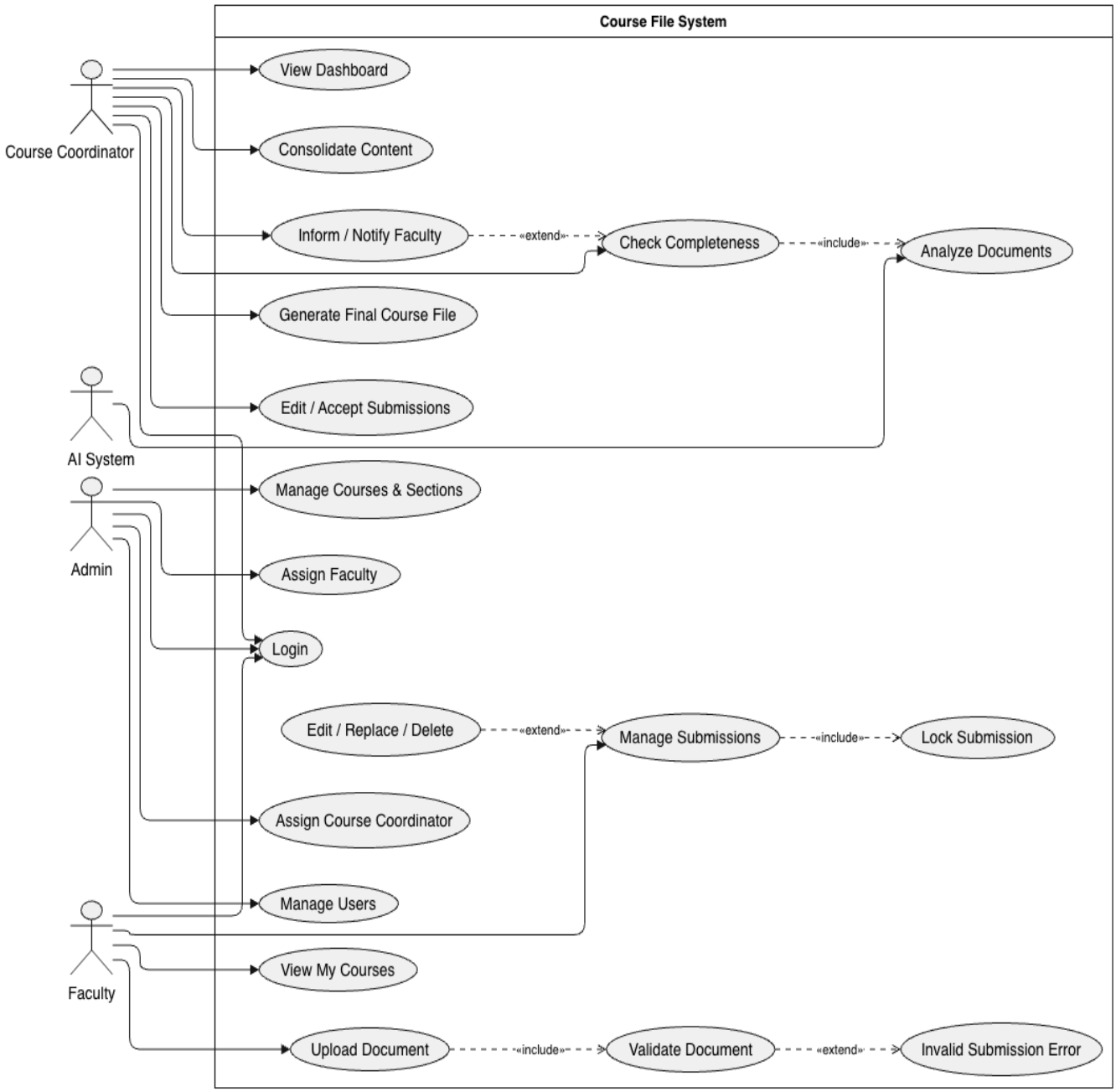
Task 1.3: Use Case Relationships

Include relationships:

- Check Completeness <<include>> Analyze Documents
- Manage Submissions <<include>> Lock Submission
- Upload Document <<include>> Validate Document

Extend relationships:

- Inform / Notify Faculty <<extend>> Check Completeness
- Edit / Replace / Delete <<extend>> Manage Submissions
- Validate Document <<extend>> Invalid Submission Error



2. Data Flow Diagram (DFD)

Task 2.1: Entities, Processes & Data Stores

External entities:

- Administrator
- Faculty
- Course Coordinator

Processes:

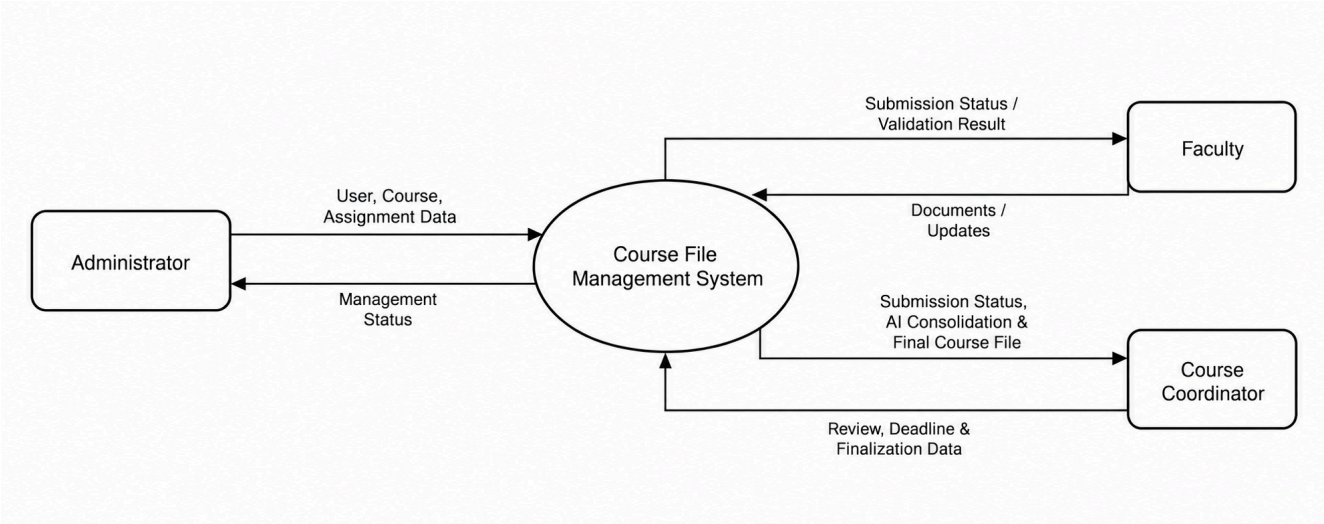
- P1 — User & Course Management
- P2 — Document Submission & Validation
- P3 — Course File Monitoring
- P4 — AI-Assisted Document Consolidation
- P5 — Review & Final Course File Generation

Data stores:

- D1 — User & Role Data
- D2 — Course & Assignment Data
- D3 — Document Repository
- D4 — Submission & Version Data
- D5 — Consolidated Course File Data

Task 2.2: Context Diagram (Level 0 DFD)

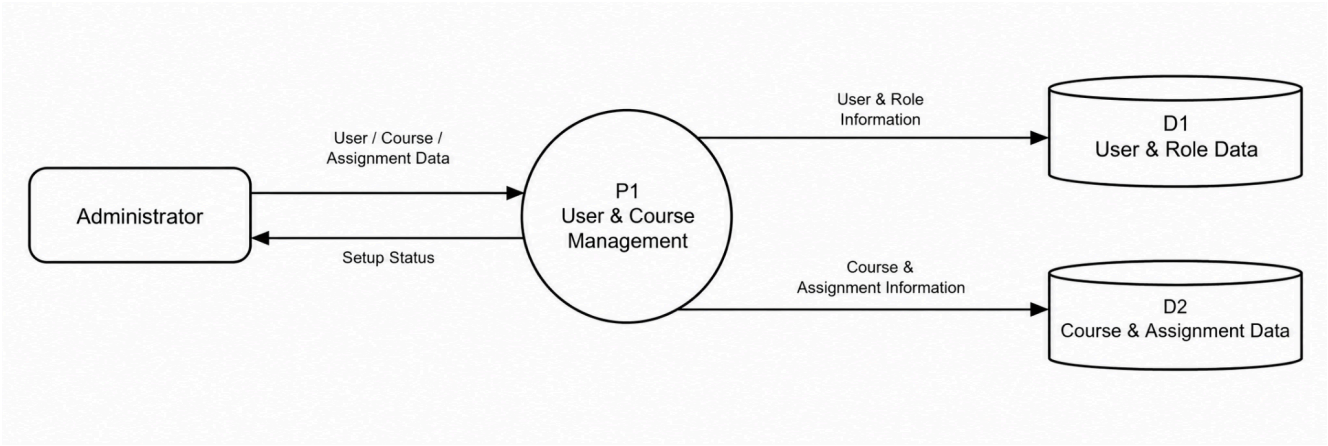
- **Administrator:** Sends User, Course, and Assignment Data; receives Management Status.
- **Faculty:** Sends Documents / Updates; receives Submission Status / Validation Result.
- **Course Coordinator:** Sends Review, Deadline & Finalization Data; receives Submission Status, AI Consolidation & Final Course File.



Task 2.3: Level 1 DFD

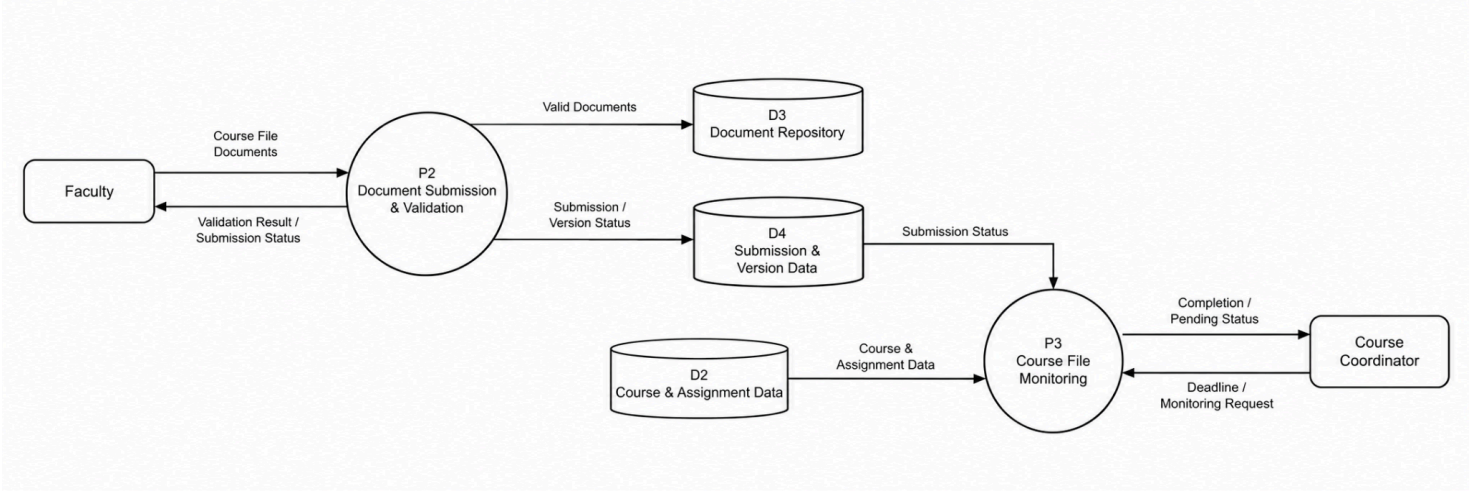
P1: User & Course Management

- Admin sends User / Course / Assignment Data to P1 and receives Setup Status.
- P1 stores User & Role Information in D1.
- P1 stores Course & Assignment Information in D2.



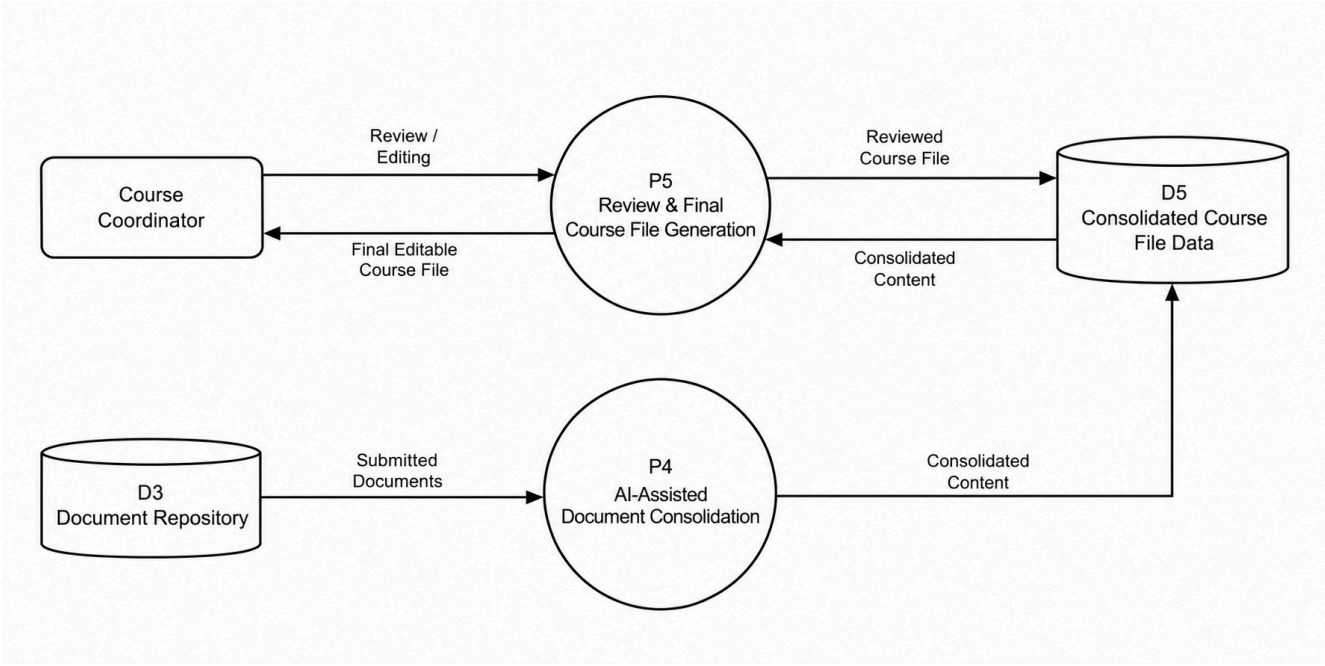
P2 & P3: Document Submission & Monitoring

- Faculty submits Course File Documents to P2 and receives Validation Result / Submission Status.
- P2 stores Valid Documents in D3 and Submission / Version Status in D4.
- P3 retrieves Submission Status from D4 and Course & Assignment Data from D2.
- Course Coordinator sends Deadline / Monitoring Request to P3 and receives Completion / Pending Status.



P4 & P5: AI Consolidation & Finalization

- D3 provides Submitted Documents to P4.
- P4 stores Consolidated Content in D5.
- P5 retrieves Consolidated Content from D5.
- Course Coordinator performs Review / Editing and receives the Final Editable Course File.
- P5 updates D5 with the Reviewed Course File.


Course Coordinator
Review / Editing
Final Editable Course File
P5
Review & Final Course File Generation
Reviewed Course File
Consolidated Content
D5
Consolidated Course File Data
D3
Document Repository
Submitted Documents
P4
AI-Assisted Document Consolidation
Consolidated Content

3. Sequence Diagram

Task 3.1: Selected Use Case Workflow

Selected workflow: End-to-End Course File System Workflow, comprising Admin Setup, Faculty Upload & Validation, and Coordinator Verification & Finalization.

Task 3.2: Objects / Lifelines

- Admin — Actor
- Faculty — Actor
- Course Coordinator — Actor
- AI System — Actor / System Component
- System / Dashboard — Boundary / Control Interface
- Database — Data Store

Task 3.3: Sequence Interactions & Steps

A. Admin Administration

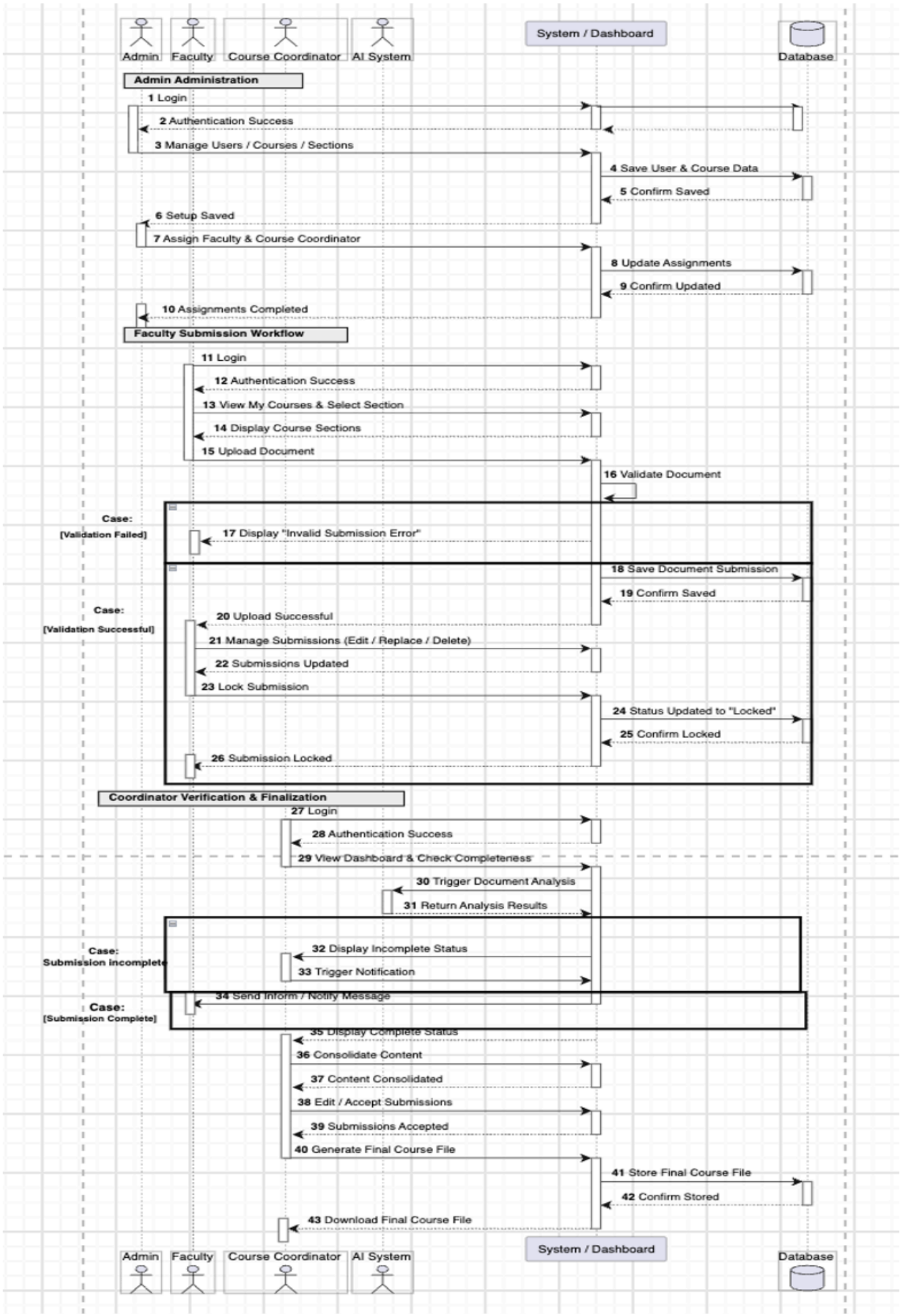
- **Steps 1–2:** Admin logs into System / Dashboard and receives Authentication Success.
- **Steps 3–4:** Admin submits Manage Users / Courses / Sections; System updates Database with Save User & Course Data.
- **Steps 5–6:** Admin triggers Assign Faculty & Course Coordinator; System performs Update Assignments in Database.

B. Faculty Submission Workflow

- **Steps 7–8:** Faculty logs into System / Dashboard and receives Authentication Success.
- **Steps 9–10:** Faculty selects View My Courses & Select Section, then performs Upload Document.
- **Step 11:** System triggers Validate Document with the AI System.
- **Step 12 — Validation Failed:** System returns an Invalid Submission Error to Faculty.
- **Step 13 — Validation Successful:** System saves the Document Submission into the Database.
- **Steps 14–16:** Faculty can Edit / Replace / Delete submissions and Lock Submission; System updates the status to Locked in the Database.

C. Coordinator Verification & Finalization

- **Steps 17–18:** Course Coordinator logs in and receives Authentication Success.
- **Step 19:** Course Coordinator selects View Dashboard & Check Completeness.
- **Steps 20–21:** System triggers Document Analysis on the AI System, which returns Analysis Results.
- **Steps 22–23 — Incomplete Branch:** System triggers a notification and sends an Inform / Notify message to Faculty.
- **Steps 24–28 — Complete Branch:** Coordinator Consolidates Content, Edits / Accepts Submissions, and Generates the Final Course File; System stores the Final Course File and provides it for download.


Admin
Faculty
Course Coordinator
AI System
System / Dashboard
Database
Admin Administration
1 Login
2 Authentication Success
3 Manage Users / Courses / Sections
4 Save User & Course Data
5 Confirm Saved
6 Setup Saved
7 Assign Faculty & Course Coordinator
8 Update Assignments
9 Confirm Updated
10 Assignments Completed
Faculty Submission Workflow
11 Login
12 Authentication Success
13 View My Courses & Select Section
14 Display Course Sections
15 Upload Document
16 Validate Document
Case:
[Validation Failed]
17 Display "Invalid Submission Error"
18 Save Document Submission
19 Confirm Saved
Case:
[Validation Successful]
20 Upload Successful
21 Manage Submissions (Edit / Replace / Delete)
22 Submissions Updated
23 Lock Submission
24 Status Updated to "Locked"
25 Confirm Locked
26 Submission Locked
Coordinator Verification & Finalization
27 Login
28 Authentication Success
29 View Dashboard & Check Completeness
30 Trigger Document Analysis
31 Return Analysis Results
Case:
Submission Incomplete
32 Display Incomplete Status
33 Trigger Notification
Case:
[Submission Complete]
34 Send Inform / Notify Message
35 Display Complete Status
36 Consolidate Content
37 Content Consolidated
38 Edit / Accept Submissions
39 Submissions Accepted
40 Generate Final Course File
41 Store Final Course File
42 Confirm Stored
43 Download Final Course File
Admin
Faculty
Course Coordinator
AI System
System / Dashboard
Database

4. State Transition Diagram

Task 4.1: Chosen Object / Entity

Course File — Primary entity tracking the academic submission lifecycle.

Task 4.2: Possible States

- Created (Course File Initialized)
- Collecting Submissions — Missing Mandatory Sections / All Mandatory Sections Submitted
- Locked for Review
- Under Coordinator Review
- Reopened for Correction
- Processing
- Draft Ready
- Under Coordinator Edit
- Content Accepted
- Ready for Generation
- Final Course File Generated
- Archived / Completed

Task 4.3 & 4.4: Events and State Transitions

1. Initial Setup

- **Course & Coordinator Assigned:** Start → Created (Course File Initialized).
- **Submission Window Opens:** Created → Collecting Submissions, beginning with Missing Mandatory Sections.

2. Submission Phase

- **Faculty Submits Missing Section:** Missing Mandatory Sections → All Mandatory Sections Submitted.
- **Deletion Creates Gap:** All Mandatory Sections Submitted → Missing Mandatory Sections.
- **Deadline Reached:** Collecting Submissions → Locked for Review.

3. Review & Reopening Phase

- **Incomplete / Invalid Submission Found:** Under Coordinator Review → Reopened for Correction.
- **Coordinator Reopens Submission:** Reopened for Correction → Collecting Submissions.
- **All Submissions Valid & Complete:** Under Coordinator Review → Processing.

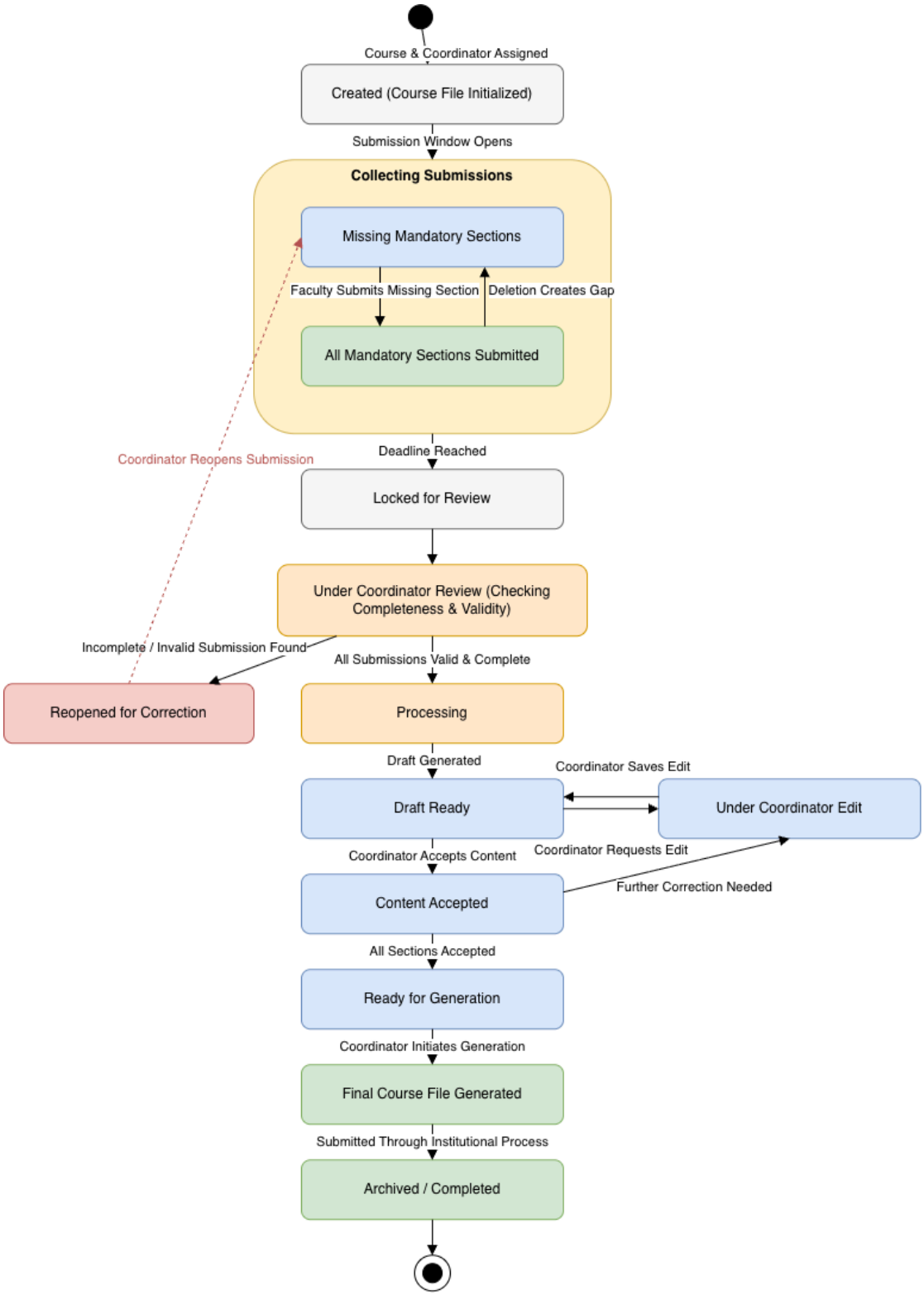
4. Draft & Edit Phase

- **Draft Generated:** Processing → Draft Ready.
- **Coordinator Requests Edit:** Draft Ready → Under Coordinator Edit.
- **Coordinator Saves Edit:** Under Coordinator Edit → Draft Ready.
- **Coordinator Accepts Content:** Draft Ready → Content Accepted.
- **Further Correction Needed:** Content Accepted → Under Coordinator Edit.

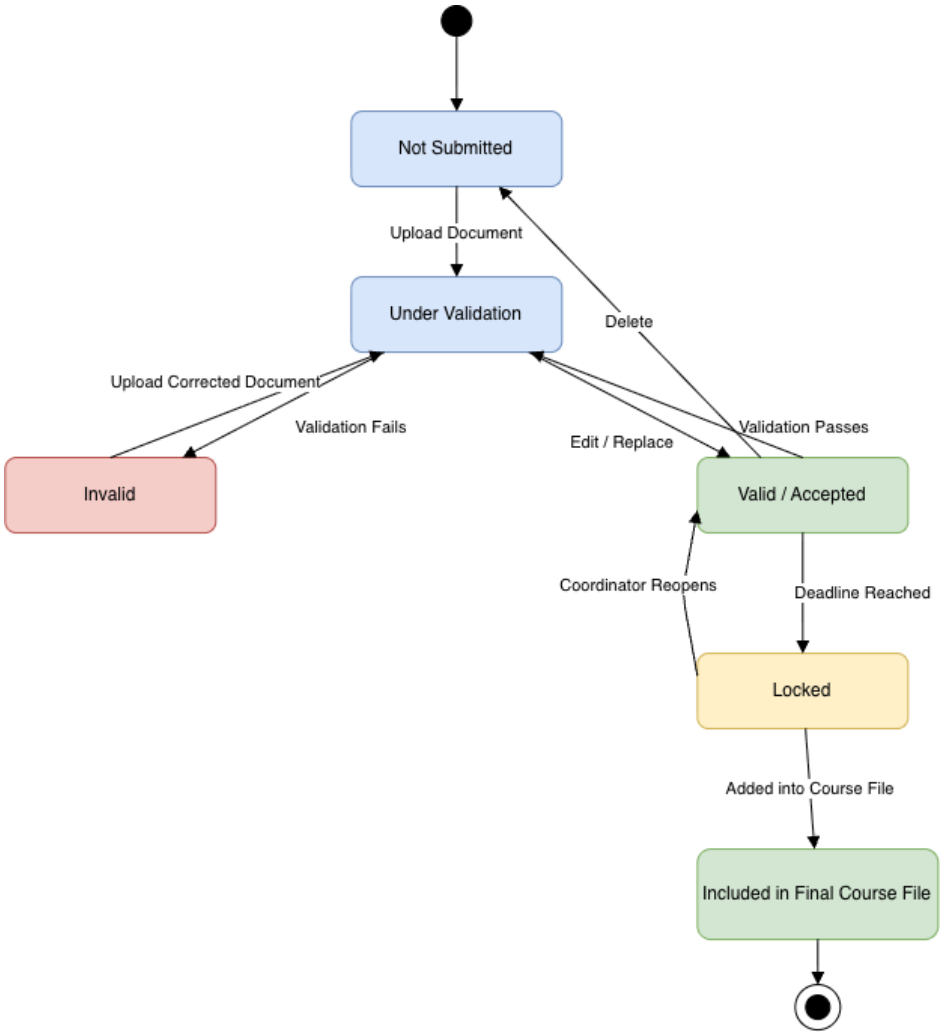
5. Finalization

- **All Sections Accepted:** Content Accepted → Ready for Generation.
- **Coordinator Initiates Generation:** Ready for Generation → Final Course File Generated.
- **Submitted Through Institutional Process:** Final Course File Generated → Archived / Completed.

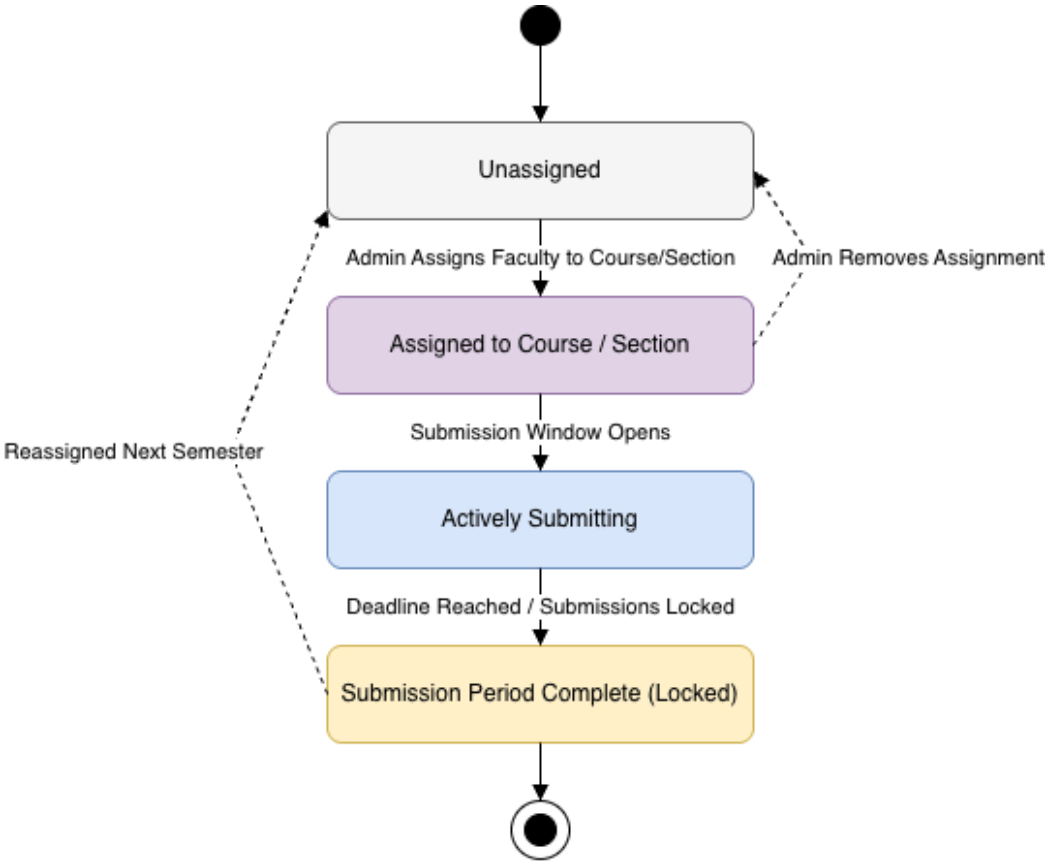
1.Course File LifeCycle



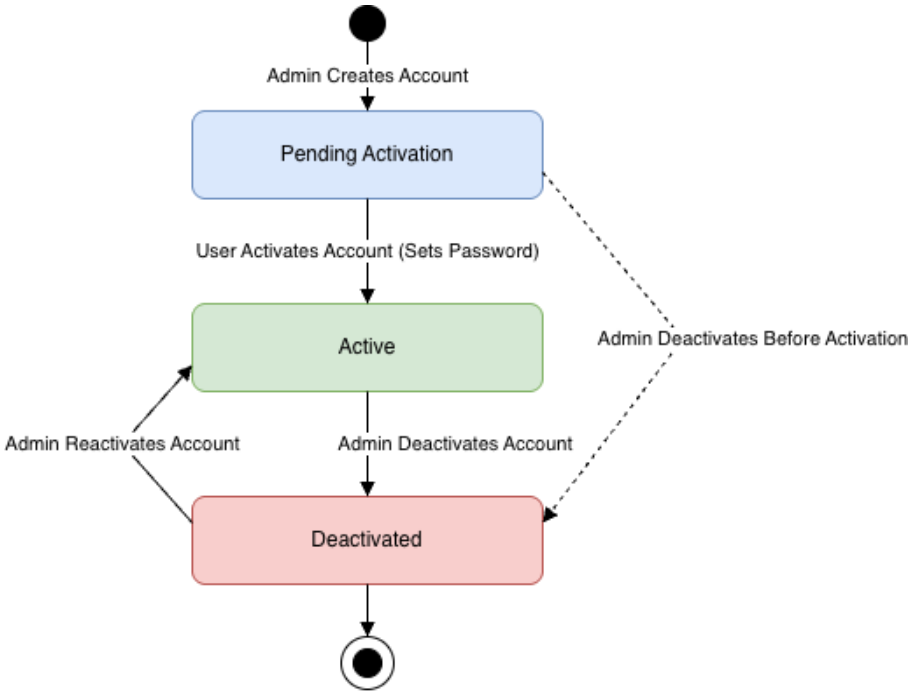
2. Submission Lifecycle



3. Faculty Assignment Lifecycle



4. User Account Lifecycle



5. Course Coordinator LifeCycle

